

Assignment 7

Name: _____

Student ID: _____

13 problems. Show all of your work.

Problem 1.

Sketch the domain of $f(x, y) = \sqrt{y - 3x}$.

Problem 2.

In Figure 1 are level curves of $g(x, y) = 2$ and $g(x, y) = 3$ for some function $g(x, y)$. What's wrong?

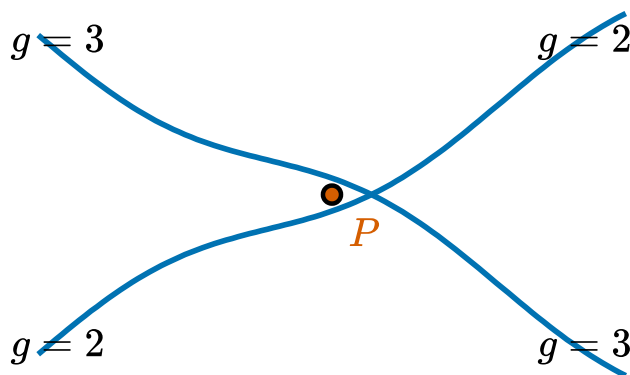


Figure 1. Two curves labelled $g = 2$ and $g = 3$ that cross each other.

Problem 3.

Let g be a function of 3 variables defined by

$$g(x, y, z) = \sqrt{4x^2 + y^2 + 9z^2}.$$

Find the equation for the level surface of g through the point $(2, \sqrt{11}, 1)$ and sketch this level surface.

Label any points where the surface crosses a coordinate axis AND label the value of g on the surface.

Problem 4.

Sketch the level surface $f(x, y, z) = z^2 - x^2 - y^2$ for $f(x, y, z) = 4$.

Problem 5. True or False

If $\lim_{(x,y) \rightarrow (a,b)} f(x,y)$ exists, the function $f(x,y)$ must be defined or continuous at (a,b) .

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Problem 6. True or False

If $f(x,y) \rightarrow L$ as $(x,y) \rightarrow (a,b)$ along any line through (a,b) , then $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Problem 7.

Discuss continuity of the following functions:

- A. $\frac{2x^4y}{x^8 + y^2}$
B. $\frac{x^2 \cos^2 x}{x^2 + (y-1)^2}$

Problem 8.

Evaluate each of the following limits or show that it does not exist.

- A. $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^5}{x^8 + y^{10}}$
B. $\lim_{(x,y) \rightarrow (1,1)} \frac{x^8 - y^8}{x^4 - y^4}$

Problem 9.

Find the limit, if it exists, or show that it does not exist:

$$\lim_{(x,y) \rightarrow (1,1)} \frac{x^2 + y^2 - 2}{|x - 1| + |y - 1|}.$$

Problem 10.

Find the following limit, if it exists, or show that it does not exist:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{y^4}{x^4 + 3y^4}.$$

Problem 11. Higher order computation

Prove that $c(x, t) = \frac{1}{\sqrt{Dt}} e^{-\frac{x^2}{4Dt}}$ is a solution of the diffusion equation

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2}.$$

Problem 12.

Let $f(x, y, z) = xy + 3y^2z^3 + \ln(xz)$. Compute $\frac{\partial^2 f}{\partial x^2}$ and $\frac{\partial^2 f}{\partial y \partial z}$.

Problem 13.

A metal plate occupies the square $0 \leq x \leq 5$, $0 \leq y \leq 5$, and is heated so that the temperature at the point (x, y) is $T(x, y)$, measured in degrees Fahrenheit. Distances are measured in feet. The table below records some measured values of T .

Measured temperatures $T(x, y)$ in °F

	$y = 1$	$y = 1.5$	$y = 2$
$x = 1$	70	65	58
$x = 1.5$	74	68	59
$x = 2$	79	75	65

- Approximate $T_x(1, 1)$ and interpret your answer, with units, in the context of the problem.
- Approximate $T_{xy}(1, 1)$, and give its units.
- An ant walks across the plate along the path

$$x = 1 - \frac{\ln(2t - 3)}{8}, \quad y = \frac{2^t}{4}, \quad 1.5 < t < 4.$$

Approximate the rate at which the temperature the ant feels is changing at the moment it is at the point $(x, y) = (1, 1)$, and give the units.